HARARE INSTITUTE OF TECHNOLOGY

SCHOOL OF INFORMATION SCIENCES AND TECHNOLOGY

B.TECH (HONS) DEGREE IN INFORMATION SECURITY AND ASSURANCE

PART 2 SEMESTER 1

ISS2103: DATA COMMUNICATIONS AND NETWORKS

TIME: 3 HOURS **DATE: NOV/ DEC 2025**

INSTRUCTIONS TO CANDIDATES:

- This question paper contains a total of **FIVE** questions.
- Each question carries **25** marks.
- Answer any **FOUR** questions.
- Start a new question on a fresh page.

ADDITIONAL MATERIAL:

- Answer booklet

Question 1

a) The OSI reference model is a fundamental concept in networking.
   i. List the layers of the OSI model starting from layer 7 in the correct order. [9]
   ii. Describe the main purpose of this model and discuss the function of the Layer 3 and Layer 4. [8]

b) With the aid of proper examples, differentiate between the Client-Server model and the Peer-to-Peer model for network applications. [8]

Question 2

a) Protocol is usually associated with the software components of the network, explain with the aid of examples what a protocol is? [5]

b) The Physical Layer is responsible for the transmission of bits over a physical medium.
   i. Explain two types of transmission media. [4]
   ii. Describe the concept of multiplexing and its importance in data communication. [4]

c) Describe any four (4) network topologies that can be used to connect devices. [12]

Question 3

a) Explain the concept of switching in networks. Differentiate between message switching, packet switching, and circuit switching. [9]

b) Compare and contrast SLAAC and DHCPv6, explaining the key differences in how they assign IPv6 addresses to network devices [6]

c) A network is experiencing a high volume of traffic, leading to delays and dropped packets. Explain the concept of congestion control and describe at least two mechanisms used to manage network congestion. [10]

Question 4

a) Given the network 172.16.0.0 with a subnet mask of 255.255.252.0, answer the following and show calculations where necessary:
   i. In which class does the network belong to? [1]
   ii. Compute the number of possible subnets. [2]
   iii. Compute the number of usable hosts are available per subnet? [2]
   iv. List the network and broadcast addresses for the first two (2) subnets. [4]
   v. Identify the valid host IP address ranges for the first three (3) subnets. [3]

b) Explain the purpose of subnetting and the benefits it provides to network administrators. [6]

c) Describe the function of a network bridge and how it operates in a Data Link Layer switching environment. [7]

Question 5

a) Describe the core principles of internetworking and expand on how TCP/IP and IP packets facilitate this process. [3]

b) Suppose a computer sends a frame to another computer on a bus topology LAN. The physical destination address of the frame is corrupted during the transmission:
   i. Explain what happens to the frame? [3]
   ii. Explain how can the sender be informed about the situation? [3]

c) Explain the difference between IP addresses and MAC addresses. Which OSI layers does each address belong to? [6]

d) Discuss the purpose of a router and a switch in a network. Which OSI layers do they operate on, and what is the difference between them in terms of functionality? [10]

.....END OF EXAMINATION PAPER.....